

LECTURE #8

Econometrics I

SELECTION OF EXPLANATORY VARIABLES & PREDICTION & QUALITATIVE INFO

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In the previous lecture #7

- ▶ We focused on data scaling:

$$\hat{y} = \hat{\beta}_0 + \boxed{\frac{\hat{\beta}_1}{c}} cx_1 + \hat{\beta}_2 x_2, \quad c\hat{y} = \boxed{c\hat{\beta}_0} + \boxed{c\hat{\beta}_1} x_1 + \boxed{c\hat{\beta}_2} x_2,$$

- ▶ variables in **logs** rescaled: **slopes not affected**, only the intercept changes.
- ▶ More on **log-level**, **level-log** functional forms, **quadratics**.
- ▶ We introduced models with **interaction terms**.
- ▶ We defined the adjusted $\bar{R}^2$ that controls for the number of

explanatory variables:

$$\boxed{\bar{R}^2 = 1 - \frac{\frac{SSR}{n-k-1}}{\frac{SST}{n-1}}}.$$

- ▶ Readings for lecture #8:
 - ▶ Chapter 6: 6.4, Chapter 7: 7.1–7.3.

Outline

Selection of independent variables

Prediction and residual analysis

Multiple regression with qualitative information

- Single dummy independent variable

- Using dummy variables for multiple categories

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Controlling for too many independent variables

- ▶ 'Quest for high goodness-of-fit' can lead to an **overspecified/overfitted** model, i.e., a model with too many independent variables.
- ▶ As the adjusted $\bar{R}^2$ penalizes for additional variables, it can also serve as a signal to stop adding independent variables.
- ▶ $\bar{R}^2$ construction tells us whether the reduction in error variance is sufficient.
- ▶ However, we should always include independent variables that affect y and are uncorrelated with all other x_j .

Four important variable selection criteria

Does an explanatory variable belong to the model?

1. **Theory:** Is including a variable in the equation unambiguous and theoretically sound? Does intuition suggest that it should be included? Also, the modeling purpose is crucial:
 - ▶ prediction/explanation,
 - ▶ vs. testing a specific theoretical/empirical relationship.
2. **Omitted variable bias reduction:** Do estimated coefficients of other variables change considerably when the variable is added to the model? It is essential to avoid serious OVB.
3. **Adjusted $\bar{R}^2$:** Does the overall fit of the equation improve (enough) when the variable is added to the model?
4. **t test and F test:** Is its coefficient statistically significant in the expected direction? F test can help us when considering excluding multiple variables or for step-wise elimination.

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Predictions in the OLS cross-sectional data framework

- ▶ Up till now, we have been dealing with estimating the parameters, testing their significance, and judging the performance and quality of the model.
- ▶ What if we estimate our model on n observations and we want to estimate $\hat{y}_{n+1}$ (having $x_{n+1,1}, \dots, x_{n+1,k}$)?
- ▶ In the cross-sectional data framework, this is referred to as **predicting** (not to be confused with **forecasting**).

Predicted/expected value

- ▶ We have the estimated model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_k x_k.$$

- ▶ **Predicted value** of y_{n+1} given a new set of explanatory variables is simply $\hat{y}_{n+1}$, i.e., we have the estimates $\hat{\beta}_j$ and we have the levels of the explanatory variables $x_{n+1,1}, \dots, x_{n+1,k}$, which we simply input to the regression model above.
- ▶ However, this is only a point estimate that is subject to **sampling variation** as obtained using the OLS estimator.
- ▶ We thus need to construct confidence intervals around such estimated $\hat{y}_{n+1}$ to represent the uncertainty of the prediction.

Confidence intervals for predictions

- ▶ Let us have specific realizations of explanatory variables and label them $c_1, \dots, c_k$ for $x_1 \dots, x_k$, respectively.
- ▶ We are thus estimating a specific parameter, which can be written as

$$\theta \equiv \mathbb{E}(y|x_1 = c_1, \dots, x_k = c_k) = \beta_0 + \beta_1 c_1 + \dots + \beta_k c_k. \quad (1)$$

- ▶ Let us use the same 'trick' as we did for testing hypotheses about linear combinations of parameters: rewrite the previous equation as

$$\beta_0 = \theta - \beta_1 c_1 - \dots - \beta_k c_k,$$

plug it in $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u$, and estimate the resulting model

$$y = \theta + \beta_1(x_1 - c_1) + \dots + \beta_k(x_k - c_k) + u.$$

- ▶ The predicted value is now the intercept of the new model, i.e., not only do we get the estimate but also the corresponding **standard error** and, in turn, the **confidence interval for prediction**.

Minimal variance for predictions

- Recall the variance of intercept in the simple regression model:

$$\text{Var}(\hat{\beta}_0) = \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2}.$$

- When is it minimal? What does it imply?
- When estimating θ , our explanatory variable becomes $x - c$.
- Rewrite the variance for θ as

$$\text{Var}(\hat{\theta}) = \frac{\sigma^2 \sum (x_i - c)^2}{n \sum (x_i - c - \bar{x} + c)^2} = \frac{\sigma^2 \sum (x_i - c)^2}{n \sum (x_i - \bar{x})^2}.$$

- Now we find the first derivative of the variance with respect to c :

$$\frac{\partial \text{Var}(\hat{\theta})}{\partial c} = -2 \frac{\sigma^2 \sum (x_i - c)}{n \sum (x_i - \bar{x})^2}.$$

- Let us finally find its minimum by laying it equal to 0:

$$\frac{\partial \text{Var}(\hat{\theta})}{\partial c} = 0 \quad \Leftrightarrow \quad \sum_{i=1}^n (x_i - c) = 0 \quad \Leftrightarrow \quad c = \frac{\sum x_i}{n} = \bar{x}.$$

'Predicting' $\mathbb{E}(y|x_{n+1})$ vs. predicting y for a specific unit

- ▶ $\hat{\theta}$ (i.e., the predicted y) and more importantly its standard error and confidence intervals give the information about the **mean/average predicted value** of y for the subpopulation with a given set of realizations of explanatory variables.
- ▶ That is why there is no u in (1).
- ▶ However, to construct the confidence interval for an individual prediction for a specific unit such as a particular person or a firm, i.e., the **prediction interval**, we also need to take the uncertainty in u into consideration.

Predicting y for a specific unit

- ▶ y^0 is the value of the dependent variable for a specific unit given values $x_1^0, \dots, x_k^0$, which can be standardly written as (cf. θ)

$$y^0 = \beta_0 + \beta_1 x_1^0 + \dots \beta_k x_k^0 + u^0.$$

- ▶ **Prediction error** is defined as

$$\hat{e}^0 = y^0 - \hat{y}^0 = \beta_0 + \beta_1 x_1^0 + \dots \beta_k x_k^0 + u^0 - \hat{y}^0.$$

- ▶ By the OLS assumptions, the prediction is unbiased, i.e., $\mathbb{E}(\hat{e}^0) = 0$.
- ▶ As only u^0 and $\hat{y}^0$ are random variables, we have

$$\text{Var}(\hat{e}^0) = \text{Var}(u^0 - \hat{y}^0) = \text{Var}(u^0) + \text{Var}(\hat{y}^0) = \sigma^2 + \text{Var}(\hat{y}^0),$$

which gives us

$$se(\hat{e}^0) = \sqrt{\hat{\sigma}^2 + \text{Var}(\hat{y}^0)}.$$

- ▶ We thus need to control both for the uncertainty in predicting y^0 and for the uncertainty of the unobserved error u^0 .

A note on residual analysis

- ▶ Residuals are not only the 'estimates of the error term'.
- ▶ They can be further used for:
 - ▶ investment decisions (overvalued firm/stock/real estate),
 - ▶ assessment of value (below/above the OLS regression line),
 - ▶ discrimination (legal) cases.

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Describing qualitative information

- ▶ Many qualitative factors can be represented as binary information: YES or NO, A or B, 1 or 0.
- ▶ Such variables are usually labelled as **binary**, **zero-one**, or **dummy** variables.
- ▶ Classical examples: employment, marital status, university degree, small baby in family, etc.

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Single dummy independent variable

- ▶ Labeling the dummy variable as D , we can simply write a model as

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_{k-1} x_{k-1} + \beta_k D + u.$$

- ▶ In practice, this gives us two models:

$$D_i = 0 \quad : \quad y_i = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_{k-1} x_{i,k-1} + u_i,$$

$$D_i = 1 \quad : \quad y_i = (\beta_0 + \beta_k) + \beta_1 x_{i,1} + \dots + \beta_{k-1} x_{i,k-1} + u_i.$$

- ▶ These models differ only in the intercepts (β_0 vs. $\beta_0 + \beta_k$), i.e., other effects are controlled for and assumed to be the same for both **groups**.
- ▶ This type is thus called an **intercept dummy**.

Dummy variable trap

- ▶ When constructing a model with dummy variables, we must not have a model which has both groups included in a model with the intercept.
- ▶ Let us have two dummy variables that are complementary, employed (E) and unemployed (U): we thus have $E + U = 1$.
- ▶ Remember how the X matrix in the matrix representation of the regression model looks:

$$X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1k} & 1 & 0 \\ 1 & x_{21} & \dots & x_{2k} & 0 & 1 \\ \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ 1 & \vdots & \dots & \vdots & 1 & 0 \\ 1 & x_{n1} & \dots & x_{nk} & 0 & 1 \end{pmatrix}$$

- ▶ If we have the intercept in our model, we violate the MLR.3 assumption: this is called the **dummy variable trap**.

A base/benchmark group

- ▶ Dummy variable trap forces us to avoid one complementary group as only 'represented by the intercept'.
- ▶ Such group is referred to as the **base** or **benchmark group**.
- ▶ This gives us the interpretation of the coefficients of dummy variables as differences between the group included in the equation and the based group 'hidden' in the intercept.
- ▶ Technically, to construct a model with both complementary dummy variables is possible if we exclude the intercept. However, such specification is less useful for statistical testing and suffers from several econometric problems.

Dummy variables and logarithms

- ▶ Nothing prevents us from constructing a model with a logarithmic dependent variable such as

$$\log(y) = \beta_0 + \beta_1 x_1 + \dots + \beta_{k-1} x_{k-1} + \beta_k D + u.$$

- ▶ Interpretation here is the same as for a 'normal' independent variable, i.e., a semi-elasticity.
- ▶ Keeping other factors fixed, the dependent variable is approximately expected to be $100\beta_k\%$ higher for group $D = 1$ compared to group $D = 0$.

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- ▶ Entire framework of a single dummy independent variable extends for **multiple categories** so that we can have a general model

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_{k-d} x_{k-d} + \\ + \beta_{k-d+1} D_1 + \dots + \beta_k D_d + u,$$

where d is the number of dummy variables.

- ▶ This gives us more information about the base group as well as more options for hypotheses testing.
- ▶ Beware the dummy variable trap here as well.
- ▶ Often, it is useful to create dummy variables as a combination of various criteria for a more straightforward interpretation.

Multiple categories: Example

- ▶ Let us have two characteristics of a person:

employment: E ,

citizenship: C .

- ▶ Let us have a straightforward model $income = \beta_0 + \beta_1 E + \beta_2 C + u$.
- ▶ It, in practice, nests a model of an expected income for each group:

employed citizens $income = \beta_0 + \beta_1 + \beta_2 + u$,

unemployed citizens $income = \beta_0 + \beta_2 + u$,

employed foreigners $income = \beta_0 + \beta_1 + u$,

unemployed foreigners $income = \beta_0 + u$.

- ▶ Testing between groups can be complicated with similar specifications, so it can be easier to create one dummy for each specific group:

employed citizens $'EC'$,

unemployed citizens $'UC'$,

employed foreigners $'EF'$,

unemployed foreigners $'UF'$.

- ▶ Then, we can pick a group against which we want to test a statistical difference (using a usual t test) and specify the model accordingly:

$$income = \gamma_0 + \gamma_1 UC + \gamma_2 EF + \gamma_3 UF + v.$$

Ordinal information

- ▶ Binary specification sometimes does not suffice.
- ▶ Assume a **qualitative** variable Q with possible values 0, 1, 2, 3.
- ▶ We can either construct a model

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_{k-1} x_{k-1} + \beta_k Q + u.$$

- ▶ Or, we can decompose Q into four new dummy categories:

$D_0 = 1$	if $Q = 0$,	$D_0 = 0$ otherwise,
$D_1 = 1$	if $Q = 1$,	$D_1 = 0$ otherwise,
$D_2 = 1$	if $Q = 2$,	$D_2 = 0$ otherwise,
$D_3 = 1$	if $Q = 3$,	$D_3 = 0$ otherwise,

and construct a model

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_{k-3} x_{k-3} + \beta_{k-2} D_1 + \beta_{k-1} D_2 + \beta_k D_3 + u.$$

- ▶ What are the practical differences between these two approaches?

Seminars and the next lecture

- ▶ Seminars:
 - ▶ practicing interaction terms,
 - ▶ model comparison,
 - ▶ predictions & CIs,
 - ▶ interpreting single dummy independent variable.
- ▶ Next lecture #9:
 - ▶ interactions involving dummy variables (slope dummies),
 - ▶ binary dependent variable: LPM,
 - ▶ more on interpreting discrete dependent variables,
- ▶ Readings for lecture #9:
 - ▶ Chapter 7: 7.4–7.7.